Getting the Surface Right: Scale-Corrected Monocular Geometry Priors for Endoscopic 4D Gaussian Splatting

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***Abstract*—Monocular 4D Gaussian splatting reconstructs deforming surgical scenes in real time, but is judged almost exclusively by how well it re-renders held-out frames, not by whether its depth is correct or where it fails. We revisit Endo-4DGS on two StereoMIS sequences and compare rendered depth with stereo-reference depth. The released preprocessing renders sequence P1 with a 34.2° instead of a 58.7° field of view, and MoGe-2, a recent metric point-map model, overestimates endoscopic depth 6–18×. On P1, correcting the intrinsics, scaling MoGe-2 from the known camera poses and inter-frame matches, and using it only for initialisation raises PSNR from 31.98 to 32.40 dB and reduces the per-frame scale-aligned depth error (AbsRel) by 37 %, from 0.234 to 0.148, over three seeds, with a higher LPIPS (0.189 vs. 0.178); absolute depth remains about 1.9× too shallow. On a static second sequence, MoGe-2 initialisation reduces the aligned error by 28 % relative to the FoV-corrected baseline. To explain where depth disagrees with the reference, we evaluate five per-pixel signals computed from the model and its prior. The accumulated opacity that Endo-4DGS uses as confidence ranks depth errors weakly (≤17 % of the attainable ranking gain), whereas disagreement with the monocular prior is the strongest tested signal (34–65 %), also under a strict left–right-consistent reference. Depth edges have the highest per-pixel error, but most of the total error lies on ordinary tissue. Image-only training from Depth-Anything gives the highest PSNR but a 61 % larger depth error than the reproduced baseline, so image-metric and depth rankings can disagree.**

***Index Terms*—Gaussian splatting, endoscopy, monocular depth estimation, deformable tissue reconstruction, surgical scene reconstruction**

I. Introduction

Reconstructing the deforming surgical field in 3D from endoscopic video is a building block for intraoperative navigation, augmented reality and autonomous robotic assistance. The task is difficult: tissue deforms continuously, instruments occlude it, specular highlights violate photometric assumptions, and the camera often moves very little. Neural radiance field methods first made such reconstructions feasible, from EndoNeRF [1] to the surface-based EndoSurf [2] and the plane-factorised LerPlane and ForPlane [3], [4], but they need minutes to hours of training and render at only a few frames per second. 3D Gaussian splatting [5] and its dynamic extensions [6] changed this, and a fast-growing family of endoscopic variants now trains in minutes and renders at 100 FPS or more, including EndoGaussian [7], EndoGS [8], Deform3DGS [9], SurgicalGaussian [10], SurgicalGS [11], Endo-4DGS [12] and, more recently, EH-SurGS [13], Endo-4DGX [14], Endo-PairGS [15], Endo-G^{2}T [16] and EndoWave [17].

When only a monocular endoscope is available, these methods rely on a learned depth prior to place and constrain the Gaussians. Endo-4DGS [12], for example, back-projects Depth-Anything [18] predictions into its initial point cloud and supervises rendered depth with them. Monocular geometry models have since advanced from relative depth to metric point maps: MoGe [19] recovers affine-invariant geometry, and MoGe-2 [20] adds metric scale and sharper detail, which makes it an attractive drop-in prior. Whether such priors actually improve the reconstructed surface, however, is hard to tell from the literature. Almost all methods are evaluated with PSNR, SSIM and LPIPS on held-out frames of a nearly static video, which reward plausible appearance rather than correct geometry; the few that report depth do so on different clips and against different references [11], [21]. Image metrics are also fragile across papers: on the same StereoMIS [22] clip, Endo-4DGS is reported at 32.69 dB by its authors [12] and at 32.17 and 32.19 dB by others [15], [16].

In this work we reproduce Endo-4DGS on StereoMIS and evaluate it on both axes, comparing rendered depth with stereo-reference depth from the same endoscope and repeating each configuration with three seeds. This exposes an error in the released preprocessing, which renders StereoMIS P1 with a 34.2° instead of a 58.7° field of view, and shows that the metric scale of MoGe-2 does not transfer to endoscopic images. We correct the intrinsics, estimate the prior's scale from the known camera poses and inter-frame feature matches rather than from stereo, and find that with a well-scaled MoGe-2 initialisation, removing the depth-related loss terms raises PSNR without increasing the scale-aligned depth error. We then ask which per-pixel signals of a trained model indicate where its depth is wrong.

Our contributions are:

- a multi-seed evaluation of monocular endoscopic 4D Gaussian splatting against stereo-reference depth, including metric and fixed-scale depth errors, which shows that image-metric rankings need not match depth-accuracy rankings on these clips;
- a correction of the StereoMIS camera intrinsics used by Endo-4DGS, worth $+0.35$ dB PSNR on P1 and $+1.19$ dB on P2_5 with unchanged scale-aligned depth error;

- a stereo-free scale estimate for metric monocular priors from known camera poses and inter-frame matches, with geometric filters and a sensitivity analysis, applicable when the camera moves;
- an initialisation-only use of the scaled MoGe-2 prior that improves PSNR by $+0.42$ dB and reduces the scale-aligned depth error by 37 % over the reproduced baseline on P1, with worse LPIPS (0.189 vs. 0.178); the FoV and initialisation findings are reproduced on a second sequence, whereas the objective and pose ablations are on P1 only;
- to our knowledge, the first evaluation of per-pixel geometric uncertainty signals for deformable Gaussian splatting in endoscopy against stereo-reference depth, with a strict left–right-consistent reference, which finds that the accumulated opacity used as confidence by Endo-4DGS ranks depth errors weakly and that disagreement with the monocular prior is the strongest tested individual signal, and attributes the remaining error to image regions.

Our changes target the depth prior and the training objective. We expect them to be complementary to the architectural advances of recent methods [15]–[17], which report higher PSNR on the same clip but no geometric accuracy; this compatibility is a hypothesis we have not tested.

II. Related Work

A. Neural fields for deformable tissue

EndoNeRF [1] first applied neural radiance fields to deformable surgical scenes, using stereo depth and tool masks with a canonical field and a deformation field. EndoSurf [2] replaced density with a signed distance function to recover explicit surfaces from stereo video, and is one of the few methods that reports geometric error (RMSE and point-cloud distance on the EndoNeRF data). LerPlane [3] and For-Plane [4] factorise space–time into planes, cutting training from hours to about ten minutes, but rendering stays below 2 FPS [9], [12].

B. Gaussian splatting for surgical scenes

3D Gaussian splatting [5] and its deformable extension 4DGS [6] brought real-time rendering, and endoscopic variants followed quickly. EndoGaussian [7] and EndoGS [8] adapted deformable Gaussians to endoscopic video; Deform3DGS [9] uses a flexible deformation model for fast training; SurgicalGaussian [10] and SurgicalGS [11] target high-fidelity reconstruction with depth from stereo networks; HFGS [23] emphasises spatial and temporal high frequencies; ST-Endo4DGS [24] models spatio-temporal structure; and EH-SurGS [13] targets efficient, high-fidelity reconstruction. Endo-4DGS [12], the basis of this work, is a monocular 4DGS method that draws its geometry prior from Depth-Anything; Endo-4DGX [14], from the same group, adds illumination correction. The most recent methods improve the representation or its priors further: Endo-PairGS [15] uses pairwise geometry priors from a feed-forward model, Endo-G^2T [16] adds geometry guidance and temporal awareness, EndoWave [17] builds on rational-wavelet 4D Gaussians, Endo-TSR [25] models temporal spectra of appearance and motion, and ColorGS [26] enriches the colour model. Surface-oriented variants use 2D Gaussians or surfels for better geometric consistency (Diff2DGS [21], Endo2DGS [27], Surgical Gaussian Surfels [28]). Other works relax the input assumptions: Free-SurGS [29] and Free-DyGS [30] remove the need for SfM or camera poses, Endo-GSMT [31] couples reconstruction with motion tracking, EndoSparse [32] handles sparse views, and Local-EndoGS [33] handles arbitrary camera motion with local deformable models.

C. Depth priors and their scale

Monocular methods depend on a learned depth prior. Depth-Anything [18] gives relative depth; MoGe [19] predicts affine-invariant point maps; MoGe-2 [20] adds metric scale and sharper detail. Because relative or foreign-domain depth carries no reliable scale, methods resolve it in different ways. Stereo-based methods sidestep the problem with depth from stereo networks [10], [11], [25]. Endo-4DGS [12] uses a scale-invariant depth loss but initialises from the prior's raw scale. Local-EndoGS [33] fits a per-frame scale and offset by least squares to the depth rendered from its own Gaussians, given known poses, and Khan *et al.* [34] optimise pseudo-metric depth with L-BFGS-B for volumetric (TSDF) reconstruction. We are not aware of prior work that uses MoGe or MoGe-2 for endoscopic Gaussian splatting, or that measures how well its metric scale transfers to endoscopic images.

D. Evaluation practice

Nearly all methods above are ranked by PSNR, SSIM and LPIPS on held-out frames of the training video. Geometric accuracy is reported by few: EndoSurf (RMSE, point-cloud distance) [2], SurgicalGS (AbsRel and related errors against RAFT stereo depth) [11], Diff2DGS (RMSE) [21], Endo2DGS (depth error on EndoNeRF) [27] and Local-EndoGS (median-scaled AbsRel and δ on two long StereoMIS sequences) [33]. Protocols also differ widely: StereoMIS is evaluated on P1 frames 800–1000 [12], [15]–[17], on clips of P2_7 and P3 [9], [21], [26], on other P1 clips [13], [14], on unnamed fast-motion clips [10], and with pose-estimation or flow-aligned metrics [30], [31]. Even on one clip, the same method differs by about 0.5 dB between papers [12], [15], [16], and among the works we examined only Endo-TSR [25], EH-SurGS [13] and this work report repeated runs.

E. Uncertainty in radiance-field reconstruction

Uncertainty for neural radiance fields and Gaussian splatting is evaluated against different targets. Several works rank *depth* error with sparsification curves (AUSE) on static, non-medical scenes: CF-NeRF [35] models a distribution over radiance fields, Bayes' Rays [36] adds post-hoc spatial uncertainty to a trained NeRF, FisherRF [37] uses the Fisher information of Gaussian parameters, also for active view selection, and training-free estimators derive it from rendering moments [38], which includes the per-ray depth variance we also test, or from

TABLE I

DESIGN COMPARISON WITH PRIOR WORK (LITERATURE REVIEWED UP TO 2 OCTOBER 2026). "–": NOT USED OR NOT REPORTED IN THE TABLES WE EXAMINED. GEOMETRY: DEPTH OR SURFACE ACCURACY REPORTED BY THE PAPER ITSELF. DA: DEPTH-ANYTHING.

Method	Representation	Depth input	Scale / pose handling	Geometry reported	StereoMIS protocol
EndoNeRF [1]	NeRF + deformation field	stereo	–	–	(run by others on P1 800–1000)
EndoSurf [2]	SDF neural field	stereo	–	RMSE, PCD (EndoNeRF data)	–
LerPlane / ForPlane [3], [4]	space–time planes	–	–	–	(run by others)
EndoGaussian [7]	deformable GS	mono or binocular	–	–	(run by others)
EndoGS [8]	deformable GS	–	–	–	(run by others)
Deform3DGS [9]	deformable GS (flexible deformation)	–	–	–	3 clips of P2_7, P3
SurgicalGaussian [10]	deformable GS	stereo (STTR-light)	–	–	2 fast-motion clips
SurgicalGS [11]	deformable GS	stereo (RAFT)	–	AbsRel, RMSE vs. RAFT depth	not stated
HFGS [23]	4DGS	–	–	–	–
Endo-4DGS [12]	4DGS	mono (DA)	scale-invariant depth loss	–	P1 800–1000
ST-Endo4DGS [24]	4DGS	–	–	–	(run by others on P1 800–1000)
Free-SurGS / Free-DyGS [29], [30]	GS	–	SfM-free / pose-free	–	– / 5 sequences
EndoSparse [32]	GS	–	sparse views	–	–
EH-SurGS [13]	deformable GS	–	–	–	P1_1, P1_2 (3 runs)
Endo-4DGX [14]	dynamic GS + illumination correction	–	–	–	two P1 clips
Endo-GSMT [31]	dynamic GS	mono	SLAM poses, motion tracking	–	5 segments, flow-aligned
Endo-G^{2}T [16]	4DGS	mono	geometry guidance	–	P1 800–1000
EndoWave [17]	rational-wavelet 4DGS	–	–	–	P1 800–1000
ColorGS [26]	deformable GS	–	–	–	P2_7, P3
Surgical Gaussian Surfels [28]	Gaussian surfels	mono (Omnidata)	–	–	P2_7, P2_8
Diff2DGS [21]	2D Gaussians	–	–	RMSE	P2_7, P3
Endo2DGS [27]	2D Gaussians, depth-guided normals	–	–	depth error (EndoNeRF data)	–
Endo-PairGS [15]	deformable GS	pairwise feed-forward priors	–	–	P1, P2-2, P2-5, P3
Endo-TSR [25]	dynamic GS	stereo (FoundationStereo)	–	–	Endo-PairGS clips (3 seeds)
Local-EndoGS [33]	local deformable GS	mono (DA)	known poses; per-frame α, β fit to rendered depth	AbsRel, δ (median-scaled)	2 sequences, 1000 / 1500 frames
Khan *et al.* [34]	TSDF fusion	mono (Depth Pro + DA)	pseudo-metric depth (L-BFGS-B)	–	– (SCARED, HEVD)
Ours	4DGS	mono (MoGe-2, metric)	P1: known poses, global scale from known-pose triangulation; P2_5: scale fitted to stereo (static camera); corrected intrinsics; prior for init only	scale-aligned, metric and fixed-scale AbsRel, δ vs. stereo SGBM	P1 801–1000, P2_5 5301–5499 (3 seeds)

multi-view warping consistency [39]. Learned combinations also exist: PRIMU [40] aggregates primitive-level error and coverage features by pixel-wise regression trained on held-out views, for colour and depth uncertainty, and reports generalisation to unseen scenes without further held-out data. Klasson *et al.* [41] compare ensembles, dropout, Laplace approximations and learned variance across sources of uncertainty. In surgery, UC-NeRF [42] uses multi-view-stereo depth uncertainty to guide sparse-view reconstruction, Endo-4DGS [12] uses a confidence map to down-weight its depth prior, and Endo-NeRF++ [43] estimates colour uncertainty for a dynamic NeRF and evaluates it against image error. In sparse-view CT, Zhao *et al.* [44] show that Gaussian-splatting variance can correlate with error globally yet fail to localise it. For monocular depth, uncertainty is benchmarked against depth error with AUSE [45], including in colonoscopy [46]. Among the works we reviewed (literature search up to 2 October 2026, covering the methods cited here), we found none that evaluates per-pixel *geometric* uncertainty of deformable Gaussian splatting in endoscopy against an independent depth reference, or that tests whether the confidence map of such a model indicates where its depth is wrong.

III. METHODOLOGY

Fig. 1 gives an overview. From a monocular endoscopic video with known camera poses, we (i) correct the camera intrinsics, (ii) predict a metric depth prior with MoGe-2 [20], (iii) estimate its scale from the known poses and inter-frame matches (on P1; on the static P2_5 clip from stereo), (iv) initialise deformable 4D Gaussians from the scaled prior, and (v) optimise them without the depth-related loss terms. The reconstruction architecture is that of Endo-4DGS [6], [12]; our changes concern the camera model, the depth prior, its scale and the training objective. Section III-I describes how we measure agreement with stereo-reference depth.

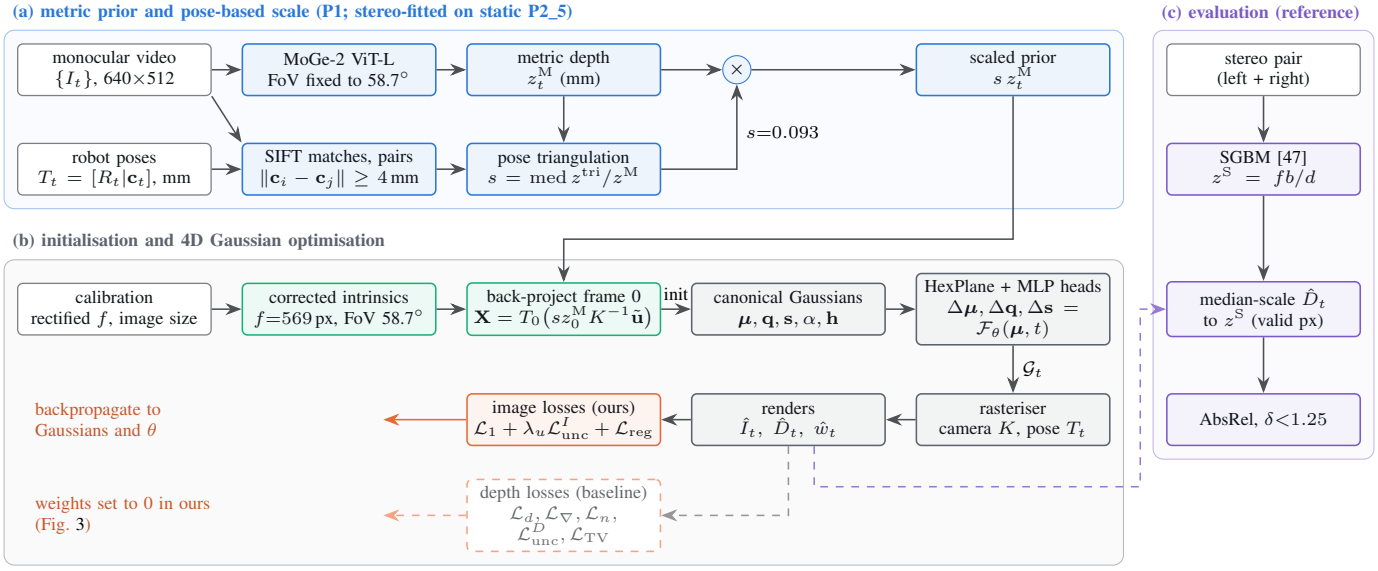


Fig. 1. Overview. (a) MoGe-2 predicts metric depth given the calibrated field of view; one global scale s is estimated by triangulating SIFT matches with the known poses (P1). On the static P2_5 clip, where triangulation has no parallax, s is instead fitted to stereo on training frames (Section V-F). (b) The scaled depth of the first frame, back-projected with the corrected intrinsics, initialises canonical Gaussians, which a HexPlane deformation field moves over time; training uses image losses only (solid), whereas the Endo-4DGS baseline also applies the depth-based losses (dashed). (c) Rendered depth is compared with stereo-reference depth; on P1 stereo is used only here.

A. Problem setting and notation

The input is a monocular video $\{I_t\}_{t=1}^{N}$ of $H{\times}W{=}512{\times}640$ rectified frames with timestamps $t/N \in [0,1)$ and camera-to-world poses $T_t = [R_t\,|\,\mathbf{c}_t]$ from the robot kinematics, in millimetres. We use the StereoMIS [22] sequence P1, frames 801–1000 (N=200). Every eighth frame, starting with the second ($(i-1) \bmod 8 = 0$ for 0-based index i; frames 802, 810, …, 994), is held out for testing, leaving 175 training frames. We distinguish two input protocols. *P1 (monocular):* the method uses the left images and the known metric poses only; the right stereo image is used only for evaluation. *P2_5 (stereo-assisted scale):* on the second, static clip (Section V-F) the prior scale is fitted to stereo depth on training frames, so stereo also enters initialisation.

B. Camera model

The renderer uses a pinhole camera with horizontal field of view $\mathrm{FoV}_x = 2\arctan\big(W/(2f)\big)$ and the principal point at the image centre. The released StereoMIS preprocessing of Endo-4DGS stores f=1039.6 px, the focal length of the unrectified 1280×1024 images, although the frames are rectified and downsampled by two. The correct value is half the rectified focal length, f=1137.99/2=569.0 px. This changes FoV_x from 34.2° to 58.7°. We use the corrected f for MoGe-2, back-projection and rendering. Two intrinsic matrices remain: as in Endo-4DGS, rendering and back-projection use K_{render} with the principal point at the image centre $(320, 256)$, whereas the scale estimation of Section III-D uses the rectified K_{tri} with principal point $(327.34, 251.33)$, i.e. offsets of $(+7.34, -4.67)$ px. MoGe-2 receives only the horizontal FoV. We did not test the effect of the residual offset on the renderer.

C. Metric geometry prior

For each frame we run MoGe-2 [20] (ViT-L with normal head, `moge-2-vitl-normal`) at its default resolution level and provide the calibrated horizontal FoV of 58.7° at inference. It returns a metric depth map z_t^{M}, a validity mask and surface normals. We convert depth to millimetres and set invalid pixels to zero; they are excluded from initialisation. Its metric scale does not transfer to these images: on the test frames the median ratio of stereo-reference depth to MoGe-2 depth is 0.168 on P1 and 0.056 on P2_5, i.e. MoGe-2 depth is 6.0× and 17.8× too large (Table V), and the ratio varies between frames (coefficient of variation 12.8 % and 6.4 %). After per-frame median alignment its shape agrees better with the stereo reference than the Depth-Anything [18] depth used by Endo-4DGS (aligned AbsRel 0.148 vs. 0.237 on P1).

D. Stereo-free scale from pose triangulation

We estimate one global factor s, a unit conversion that maps z^{M} into the metric frame of the poses (Fig. 2). This assumes that matched points move little between the two frames compared with the camera-induced parallax; deformation, breathing, instruments and reflections violate it, so we filter correspondences geometrically. For frame pairs (i, j) from a pool sampled every 5 frames, whose camera centres are at least 4 mm apart, we match SIFT features (4000 per frame, ratio test 0.75) and reject outliers with a RANSAC fundamental matrix (1 px threshold). We then keep only matches that are consistent with the *known* poses (Sampson distance ≤2 px to the fundamental matrix implied by T_i, T_j), triangulate in front

of both cameras and reproject within 2 px in both views. Each remaining match $(\mathbf{u}_k, \mathbf{u}'_k)$ is triangulated with the projection matrices

$$P_i = K_{\text{tri}}\,[\,R_i^\top \mid -R_i^\top \mathbf{c}_i\,], \qquad P_j = K_{\text{tri}}\,[\,R_j^\top \mid -R_j^\top \mathbf{c}_j\,], \tag{1}$$

which gives a depth z_k^{tri} in camera i. The scale is the median of per-pair medians,

$$s = \underset{(i,j)}{\text{median}}\ \underset{k}{\text{median}}\ \frac{z_k^{\text{tri}}}{z_i^{\text{M}}(\mathbf{u}_k)}, \tag{2}$$

where we keep only pairs with at least 20 valid ratios. The triangulated depth is computed independently of the prior, but s divides by the prior and therefore inherits its frame-to-frame scale variation.

Frame pool and sensitivity. The experiments of Sections V–VI use s=0.0928, which was estimated before we restricted the pool and added the known-pose filters: it used all 200 frames, including the 25 test frames, and only the RANSAC filter (144 pairs, inter-quartile range 0.085–0.107). With training frames only it becomes 0.0976 (116 pairs), and with the known-pose filters 0.1044 (51 pairs, 35 training frames 801, 806, …; IQR 0.087–0.108). The cheirality, reprojection and triangulation-angle ($\geq$1–2°) filters remove no further pairs. The estimate is stable for baselines of at least 4 mm (0.1044 at 4 mm, 0.1054 at 6 mm with IQR 0.105–0.108) but biased low at 2 mm (0.085, IQR 0.065–0.104). We retrain our model with the held-out-clean factor 0.1044 in Section V to check that the conclusions do not depend on the test frames. The factor differs from the stereo fit on P1 (0.165); candidate explanations are kinematic pose error (2–6 px epipolar error between robot poses and image matches), tissue motion and the different pixel sets used by the two fits, and we do not attribute the gap to one of them. A single factor converts units consistently with the poses; it does not correct the frame-to-frame scale variation of MoGe-2, and our model initialises from one frame only. Solving for s directly from back-projected prior points, $s\,(R_i P_i^{\text{M}} - R_j P_j^{\text{M}}) = \mathbf{c}_j - \mathbf{c}_i$, failed in our setting (estimates of 0.006–0.026) because the ∼10 mm error of the prior overwhelms baselines of a few millimetres.

E. Initialisation

The canonical Gaussians are created by back-projecting the scaled prior of the first training frame with the corrected intrinsics,

$$\mathbf{X}(\mathbf{u}) = R_0\left(s\, z_0^{\text{M}}(\mathbf{u})\, K^{-1}\tilde{\mathbf{u}}\right) + \mathbf{c}_0, \tag{3}$$

for every pixel $\mathbf{u}$ with valid prior depth that is not masked as an instrument, using K_{render}. Here $\tilde{\mathbf{u}}$ is the homogeneous pixel coordinate, and each point takes the pixel's colour as its initial colour. This yields at most 327 680 Gaussians for a 640×512 frame; on P1 all pixels of the first training frame (801) are valid and unmasked, so the initial count is exactly 327 680 (the final count after densification and pruning is reported separately in Table II). The baseline is initialised the same way from its Depth-Anything depth (the inverse of the predicted disparity, scaled by 10^3 and clipped to its 5th–95th percentiles) and the uncorrected intrinsics.

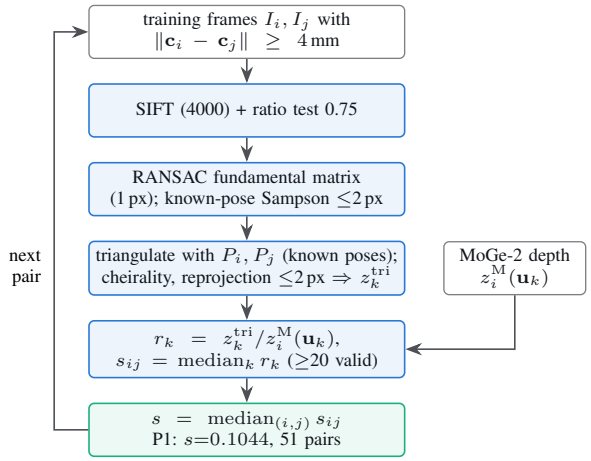


Fig. 2. Estimation of the global prior scale s from known poses and inter-frame matches (Eq. 2); stereo depth is not used. Shown with the corrected frame pool and filters; the main experiments used s=0.0928 from the earlier, unfiltered all-frame estimate (Section III-D).

F. Deformable 4D Gaussian representation

Each Gaussian k has a mean $\boldsymbol{\mu}_k$, rotation quaternion $\mathbf{q}_k$, scale $\mathbf{s}_k$, opacity α_k and spherical-harmonic colour $\mathbf{h}_k$ (degree 3), with covariance $\Sigma_k = R(\mathbf{q}_k)S(\mathbf{s}_k)S(\mathbf{s}_k)^\top R(\mathbf{q}_k)^\top$. Following Endo-4DGS and 4DGS [6], [12], a deformation field maps a canonical Gaussian to time t:

$$(\Delta\boldsymbol{\mu}_k, \Delta\mathbf{q}_k, \Delta\mathbf{s}_k) = \mathcal{F}_\theta\big(\text{HexPlane}(\boldsymbol{\mu}_k, t)\big), \tag{4}$$

where HexPlane features come from six 2D planes over (x, y, z, t) at resolution 64×64×64×75 with 32 channels and two multi-resolution levels, and $\mathcal{F}_\theta$ consists of small MLP heads of width 64. Opacity and colour are not deformed. With the compositing weights $a_k(\mathbf{p}) = \alpha'_k \prod_{l<k}(1-\alpha'_l)$ of the depth-sorted Gaussians, the rasteriser returns

$$\begin{aligned} \hat{I}(\mathbf{p}) &= \textstyle\sum_k a_k \mathbf{c}_k + \big(1 - \sum_k a_k\big)\mathbf{b}, \\ \hat{w}(\mathbf{p}) &= \textstyle\sum_k a_k, \qquad \hat{D}(\mathbf{p}) = \sum_k a_k z_k \,/\, \hat{w}(\mathbf{p}), \end{aligned} \tag{5}$$

with background colour $\mathbf{b}$ and camera-space depths z_k. $\hat{D}$ is the opacity-normalised expected depth along the ray, not necessarily the first surface intersection; pixels with $\hat{w}$=0 have no depth and are excluded from all depth metrics. Endo-4DGS uses the accumulated opacity $\hat{w}$ as a confidence map.

G. Training objective

Endo-4DGS minimises

$$\begin{aligned} \mathcal{L} = \mathcal{L}_1(\hat{I}, I) + \lambda_u \mathcal{L}^I_{\text{unc}} + \mathcal{L}_{\text{reg}} \\ + \underbrace{\lambda_d\big(\mathcal{L}_d + \mathcal{L}_\nabla + \mathcal{L}_{\text{TV}}\big) + \lambda_n \mathcal{L}_n + \lambda_u \mathcal{L}^D_{\text{unc}}}_{\text{depth-based terms}}. \end{aligned} \tag{6}$$

Here $\mathcal{L}_d$ is an L1 loss between max-normalised rendered and prior depth, $\mathcal{L}_\nabla$ a depth-gradient loss, $\mathcal{L}_{\text{TV}}$ total variation

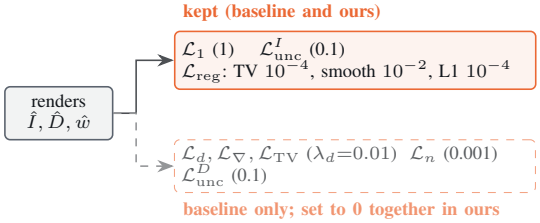


Fig. 3. Loss terms and weights (Eq. 6). Our model keeps only the image and HexPlane terms; note that λ_d also scales the image TV term.

of the rendered depth and image, and $\mathcal{L}_n$ an L1 loss between rendered normals and pseudo-normals from prior-depth gradients. The uncertainty terms are computed with global reductions over the valid pixels $\mathcal{V}$, $\mathcal{L}^I_{\text{unc}} = \big(\frac{1}{|\mathcal{V}|}\sum_{\mathbf{p}} \|\hat{I} - I\|^2\big)/\big(\frac{2}{|\mathcal{V}|}\sum_{\mathbf{p}} \hat{w}^2\big) + \frac{1}{|\mathcal{V}|}\sum_{\mathbf{p}} \log \hat{w}$, and likewise $\mathcal{L}^D_{\text{unc}}$ on max-normalised depth; a pixel's error is therefore not weighted by its own $\hat{w}$, and $\hat{w}$ is not trained as a calibrated per-pixel error estimate. Finally, $\mathcal{L}_{\text{reg}}$ regularises the HexPlanes with spatial total variation, temporal smoothness and L1 on the time planes. Fig. 3 lists the weights. The max-normalised terms $\mathcal{L}_d$ and $\mathcal{L}^D_{\text{unc}}$ are invariant to a multiplicative depth scale; the pseudo-normals (built from raw prior-depth differences, $[\partial_x z, -\partial_y z, -1]$ normalised) and the total variation of the rendered depth are evaluated in raw depth units and are not, and the image TV term shares the weight λ_d in the released code. With the scaled MoGe-2 initialisation we set λ_d, λ_n and the depth-uncertainty weight to zero, which disables the depth, depth-gradient, depth-TV and image-TV, pseudo-normal and depth-uncertainty terms *together*; Section V shows that removing this group improves PSNR, but does not isolate the effect of each term.

H. Optimisation

Training follows Endo-4DGS: 2000 coarse iterations without deformation, then 6000 fine iterations with deformation, batch size 1. Gaussians are densified and pruned every 100 iterations between iterations 500 and 3000, and opacity is never reset. Learning rates are the Endo-4DGS defaults; for example, the position rate decays from 1.6×10^{-4} to 1.6×10^{-6}, the deformation MLP rate from 1.6×10^{-4} to 1.6×10^{-6} and the HexPlane rate from 1.6×10^{-3} to 1.6×10^{-5}. One run takes about 16 min on a single RTX 4060 Ti, and the trained model renders at 38 FPS.

I. Geometric evaluation protocol

Image quality is reported as PSNR, SSIM and LPIPS (AlexNet) on the 25 test frames. No ground-truth surface is available, so we compare against a *stereo reference*: depth estimated from the rectified stereo pair with semi-global block matching [47], using 256 disparities, block size 5, smoothness penalties P_1=600 and P_2=2400, uniqueness ratio 10 and speckle filtering (window 100, range 2), and $z^{\text{S}} = fb/d$ with fb=2494.6 px·mm at 640×512. Our main geometric metric is a *shape* diagnostic: each rendered depth map $\hat{D}_t$ is multiplied by $\text{median}(z^{\text{S}}/\hat{D}_t)$ over Ω before computing

$$\text{AbsRel} = \frac{1}{|\Omega|}\sum_{\mathbf{p}\in\Omega} \frac{|\hat{D}(\mathbf{p}) - z^{\text{S}}(\mathbf{p})|}{z^{\text{S}}(\mathbf{p})}, \tag{7}$$

$$\delta = \frac{1}{|\Omega|}\sum_{\mathbf{p}\in\Omega} \Big[\max\Big(\frac{\hat{D}(\mathbf{p})}{z^{\text{S}}(\mathbf{p})}, \frac{z^{\text{S}}(\mathbf{p})}{\hat{D}(\mathbf{p})}\Big) < 1.25\Big], \tag{8}$$

where Ω is the set of pixels with SGBM disparity > 1 px (valid stereo depth; ≈50 % of each frame, the leftmost 256 columns being outside the disparity search range) and finite, positive rendered depth. We call this the *per-frame scale-aligned, stereo-reference-relative* AbsRel. Per-frame alignment removes absolute and frame-to-frame scale errors, which matter because the poses are metric. We therefore also report the *metric* AbsRel of the raw rendered depth, the AbsRel after *one* scale factor fitted on the training views, and the statistics of the per-frame factors (Table VI). Metrics are averaged over the test frames and then over three training seeds; ± denotes the sample standard deviation over seeds ($n-1$).

J. Variants examined

MoGe-2 normal supervision. We replace $\mathcal{L}_n$ by a cosine loss between rendered and MoGe-2 normals. Each Gaussian's normal (the axis of its smallest scale) is rotated into the camera frame, flipped to face the camera and composited on a black background. This removes the view-dependent colour terms and the white-background bias that affect the baseline's normal map. The loss is $\lambda\,(1 - \langle\hat{\mathbf{n}}, \mathbf{n}^{\text{M}}\rangle)$, averaged over valid pixels, with $\lambda \in \{0.01, 0.05\}$.

Bundle-adjusted poses. Starting from the robot poses, we triangulate exhaustive SIFT matches and run COLMAP [48] bundle adjustment with fixed intrinsics. We align the result back to the robot frame: the rotation is the chordal mean of the per-camera offsets, and the scale and offset come from a least-squares fit of the camera centres. Optionally, only a temporally smoothed correction is applied (a Gaussian filter with σ=15 frames on the per-frame rotation vector and translation offset).

K. Explaining geometric error: uncertainty signals

To explain *where* a reconstruction disagrees with the stereo reference, we compare per-pixel signals that are computed from a trained model and its monocular inputs only, and score them against the per-frame scale-aligned, stereo-reference-relative error $e(\mathbf{p}) = |s_t\hat{D}(\mathbf{p}) - z^{\text{S}}(\mathbf{p})|/z^{\text{S}}(\mathbf{p})$ of Section III-I. All signal maps are computed and frozen *before* any stereo-derived mask is applied; the stereo reference only selects the pixels on which they are scored and defines the target.

- *Accumulated opacity* $\hat{w}$, which Endo-4DGS uses as its confidence map (Section III-G); uncertainty $u_{\text{cov}} = 1 - \hat{w}$.
- *Ray depth spread.* With the compositing weights a_i, the normalised raw moments $m_r = \sum_i a_i z_i^r/\hat{w}$ for $r \in \{1, 2\}$ give $\sigma_z = \sqrt{\max(m_2 - m_1^2, 0)}$ and $u_{\text{ray}} = \sigma_z/\max(\hat{D}, \epsilon)$ with $\epsilon = 10^{-6}$; m_1, m_2 are obtained in one extra rasterisation pass that renders (z_i, z_i^2) as colours on a black background.

- *Deformation magnitude* $u_{\mathrm{mot}} = \big(\sum_i a_i \|\Delta\boldsymbol{\mu}_i\|/\hat{w}\big)/\max(\hat{D},\epsilon)$ at the frame's time.
- *Prior disagreement* $u_{\mathrm{prior}} = |\hat{D} - \tilde{z}|/\max(\hat{D},\epsilon)$, where $\tilde{z} = c\, z^{\mathrm{prior}}$ is the monocular prior of the same frame (Depth-Anything or MoGe-2, whichever the model was built from) and $c = \mathrm{median}(\hat{D}/z^{\mathrm{prior}})$ over pixels where both are finite and positive.
- *Seed ensemble* u_{ens}: the relative standard deviation of the rendered depth of three models trained with different seeds, each normalised by its own median over the image.

We also test a combination without fitted parameters, the mean of the per-frame percentile ranks of u_{ray}, u_{prior} and u_{ens}, which unlike learned predictors [40] needs no held-out error.

Metrics. Signals are scored on $\Omega \cap \{z^{\mathrm{prior}} > 0\}$. For each test frame and seed we sort pixels by decreasing uncertainty, remove the most uncertain fraction f for 50 values $f \in \{0, 0.02, \ldots, 0.98\}$ and record the mean AbsRel of the remaining pixels; the oracle curve sorts by the true error. AUSE is the trapezoidal area between a signal's curve and the oracle curve, and AURG the area between the random-order curve (the constant mean error) and the signal's curve [45]; both are unnormalised AbsRel integrals. We summarise each signal by the share of the attainable ranking gain,

$$G = 100\,\frac{\mathrm{mean}_{t,s}\,\mathrm{AURG}_{t,s}}{\mathrm{mean}_{t,s}\,\mathrm{AUSE}^{\mathrm{random}}_{t,s}}, \tag{9}$$

a ratio of means over frames t and seeds s, where $\mathrm{AUSE}^{\mathrm{random}}$ is the area between the random and oracle curves; $G{=}100\,\%$ matches the oracle and $G{=}0$ a random order. The mean of per-frame ratios differs from G by at most 5.2 percentage points over all signals, setups and references and leads to the same conclusions. We also report Spearman's ρ between u and e (ties by average rank), and the mean error per uncertainty decile, a ranking diagnostic rather than a calibration of predicted error magnitudes.

Strict reference. SGBM depth is itself an estimate. We therefore repeat every evaluation with a strict reference that keeps only pixels whose left- and right-view disparities agree within 1 px (the right-view disparity is computed on the mirrored pair), retaining 20–25 % of the pixels. This reduces sensitivity to some stereo mismatches, but mutually consistent wrong matches remain and the evaluated pixels change.

Failure attribution. We partition the scored pixels into four disjoint regions with fixed precedence: the blue ring-shaped device visible in P1 (OpenCV hue 80–130, saturation > 0.35, value > 0.15, opened with a 5×5 kernel), then specular highlights (value > 0.9 and saturation < 0.3, dilated with a 5×5 kernel), then depth edges (top 10 % of $\|\nabla z^{\mathrm{S}}\|/z^{\mathrm{S}}$ over valid pixels, with 3×3 Sobel gradients after Gaussian smoothing with $\sigma{=}1.5$ px; invalid stereo pixels are filled with the median depth before smoothing, which can create spurious edges at mask borders), and the remaining tissue. Edge detection uses the reference, so attribution is an analysis tool rather than a test-time signal. We report the mean AbsRel per region and each region's share of the total error, computed from per-frame error sums.

TABLE II
HARDWARE, SOFTWARE AND COMPUTE COST.

Hardware: RTX 4060 Ti 16 GB, i9-12900K (24 threads), 62 GB RAM, Ubuntu 24.04
Software: Py 3.8/PyTorch 2.0.0/CUDA 11.8 (Endo-4DGS); Py 3.10/PyTorch 2.4.1 (MoGe-2); COLMAP 4.2

Depth prior	Params	Device	Time / frame
Depth-Anything ViT-B/14 (ONNX)	97 M	CPU	0.75 s
MoGe-2 ViT-L (3.2 GB peak)	331 M	GPU	0.22 s
Trained model	**Gaussians**	**Render**	**Train (8k it.)**
Original [12]	330 k	38 FPS	16.0 min
Corrected FoV + DA	352 k	35 FPS	17.0 min
Ours (FoV + MoGe-2 init)	327 k	38 FPS	15.9 min

IV. Experimental Setup

Data and protocol. The main experiments use StereoMIS [22] sequence P1, frames 801–1000, with the split, metrics and geometric evaluation of Section III-I, under the monocular input protocol (left images and known poses). Section V-F repeats the FoV and initialisation comparisons on a second sequence, P2_5, where a separate stereo-assisted scale protocol is also reported, and Section VI evaluates uncertainty on both. Each main configuration is trained with seeds 0, 1 and 2; we report the mean and the sample standard deviation over seeds $(n{-}1)$. Ablations use one seed unless marked otherwise; three runs on two short, temporally correlated clips do not support significance claims, and we make none.

Reproduction. We use the released Endo-4DGS code (commit 39c6ce2) with its StereoMIS configuration. It differs from the description in the paper [12]: it uses Depth-Anything ViT-B/14 (ONNX), whereas the paper describes Depth-Anything Small, and the loss weights are those of the released configuration. Our local modifications are the depth-prior selector, the optional normal loss, the seed option and the evaluation scripts; with the default options the training code path is unchanged. We therefore call our baseline a reproduction of the released code, not of the paper.

Compared configurations. (1) *Original*: released Endo-4DGS with Depth-Anything (DA) depth and the 34.2° field of view. (2) + *FoV fix*: the same with corrected intrinsics (Section III-B). (3) + *FoV, DA init only*: corrected intrinsics, DA initialisation and the depth-related terms disabled. (4) + *FoV, MoGe-2 + depth loss*: pose-scaled MoGe-2 (×0.093) for initialisation and as the depth target of the full objective. (5) *Ours*: corrected intrinsics, pose-scaled MoGe-2 initialisation (×0.093) and the depth-related terms disabled. (6) *Ours, held-out-clean scale*: as (5) with ×0.104, estimated from training frames with the known-pose filters (Section III-D).

Hardware and software. Table II lists the platform and the cost of each component. Endo-4DGS and the evaluation run in one environment; MoGe-2 runs in a separate one because it requires a newer Python.

V. Results

A. Quantitative comparison

Fixing the field of view adds 0.35 dB PSNR for DA but leaves its scale-aligned depth error unchanged (AbsRel 0.235 vs. 0.234, with a large seed spread of 0.039; Table III). MoGe-2 with the default depth losses gives a much lower depth

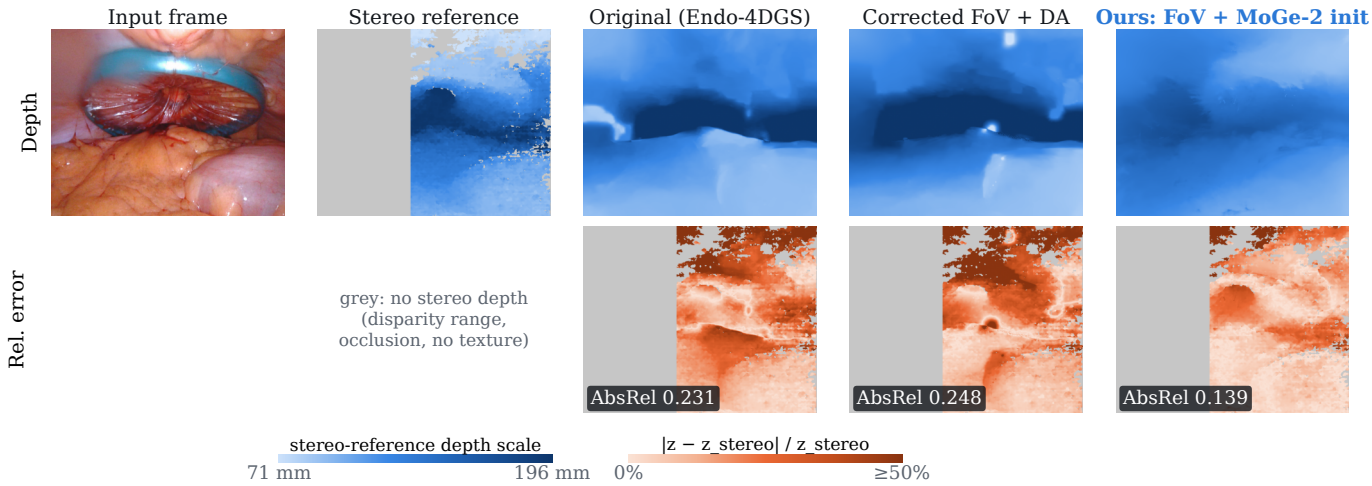


Fig. 4. Qualitative comparison on test frame 898 (median-error seed of each setup). Top: rendered depth on a shared scale; bottom: relative error against stereo depth. Correcting the field of view alone leaves the Depth-Anything geometry unchanged; MoGe-2 initialisation removes the spurious depth step below the instrument ring and lowers AbsRel from 0.231 to 0.139.

TABLE III
MAIN RESULTS ON STEREOMIS P1 (MEAN ± SAMPLE S.D., 3 SEEDS). ABSREL AND δ: PER-FRAME SCALE-ALIGNED, AGAINST SGBM STEREO; SEE TABLE VI FOR METRIC AND FIXED-SCALE ERRORS.

Setup	PSNR↑	SSIM↑	LPIPS↓	AbsRel↓	δ<1.25 ↑
Reported [12]†	*32.69*	*0.850*	*0.148*	–	–
Original, reproduced	31.98±.09	0.8515±.0010	0.178±.003	0.234±.004	0.603±.014
+ FoV fix (DA)	32.33±.14	0.8536±.0019	0.186±.002	0.235±.039	0.611±.067
+ FoV, DA init only‡	32.62±.13	0.8598±.0016	0.166±.001	0.376±.048	0.404±.078
+ FoV, MoGe-2 + depth loss	31.87±.13	0.8436±.0017	0.216±.004	0.155±.003	0.778±.008
Ours: FoV, MoGe-2 init (×0.093)	32.40±.10	0.8526±.0013	0.189±.002	0.148±.001	0.795±.001
Ours, held-out-clean scale (×0.104)	32.29±.06	0.8506±.0007	0.194±.001	0.147±.001	0.799±.003

†As printed in [12] (arXiv v4, Table 2; frames 800–1000; Depth-Anything Small, the paper's loss weights). Our reproduction uses the released code (commit 39c6ce2; Depth-Anything ViT-B). ‡Depth-related terms and image TV disabled, as for ours.

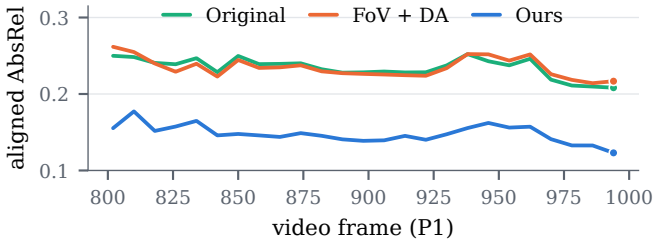


Fig. 5. Rendered-depth error per test frame (mean of 3 seeds).

TABLE IV
ABLATIONS ON P1 WITH CORRECTED FOV (1 SEED; *3-SEED MEAN). EACH BLOCK VARIES ONE FACTOR AT THE PRIOR SCALE GIVEN IN ITS FIRST COLUMN; ROWS OF DIFFERENT BLOCKS ARE NOT DIRECTLY COMPARABLE. ABSREL: PER-FRAME SCALE-ALIGNED, AGAINST STEREO.

Factor	Variant	PSNR↑	AbsRel↓
MoGe-2 scale	raw metric (×1), 34.2° FoV	29.51	0.186
	stereo-fitted (×0.165), 34.2° FoV*	31.09	0.162
	stereo-fitted (×0.165), 58.7° FoV	31.56	0.156
	pose-triangulated (×0.093), 58.7° FoV*	31.87	0.155
Depth weight λ_d	0.03	30.99	0.159
(pose-scaled MoGe-2, ×0.093)	0.01 (default)*	31.87	0.155
	0.003	32.13	0.148
	all depth terms and image TV off ($\lambda_d{=}\lambda_n{=}\lambda_u^D{=}0$)*	**32.40**	**0.148**
Normal loss	reference: pseudo-normals, w=0.001	31.56	0.156
(stereo-scaled MoGe-2, ×0.165)	MoGe-2 normals, w=0.01 / 0.05	31.44 / 31.28	0.155 / 0.153
Poses‡	raw BA: DA / MoGe-2 init	29.97 / 30.73	0.292 / 0.149
	smoothed BA: DA / MoGe-2 init	32.08 / 31.98	0.255 / 0.149
	robot poses: DA* / MoGe-2 init*	32.33 / 32.40	0.235 / 0.148

‡The MoGe-2 rows with BA poses re-estimate the prior scale from those poses (×0.123 raw, ×0.124 smoothed), so pose and scale change together.

error (0.155) but 0.46 dB lower PSNR than FoV-corrected DA. Removing the group of depth-related terms (Section III-G) recovers the PSNR: our model is 0.42 dB above the reproduced baseline in PSNR (+0.37 to +0.45 dB per seed) and reduces the scale-aligned AbsRel by 37 % relative to it (seed s.d. 0.0004), with an SSIM change of only +0.001 and a worse LPIPS (0.189 vs. 0.178). The improvement is therefore in PSNR and depth agreement, not in every appearance metric. Against FoV-corrected DA the PSNR difference is small (+0.07 dB) and the gain is in depth agreement. The same objective change applied to DA gives the highest PSNR (32.62 dB) but a 61 % larger scale-aligned depth error than the reproduced baseline (0.376 vs. 0.234; 154 % larger than ours), so a better image metric does not imply better depth agreement (Fig. 6). With the scale factor re-estimated from training frames only and filtered for consistency with the known poses (×0.104; Section III-D), the model reaches 32.29 dB and 0.147 AbsRel, i.e. the same depth agreement and a 0.11 dB smaller PSNR gain. Our model has the lowest error on all 25 test frames (Fig. 5) at comparable training time and rendering speed (Table II).

B. Ablations

Each block varies one factor at the prior scale stated in the table. Raw MoGe-2 depth must be rescaled (29.51 dB), and at

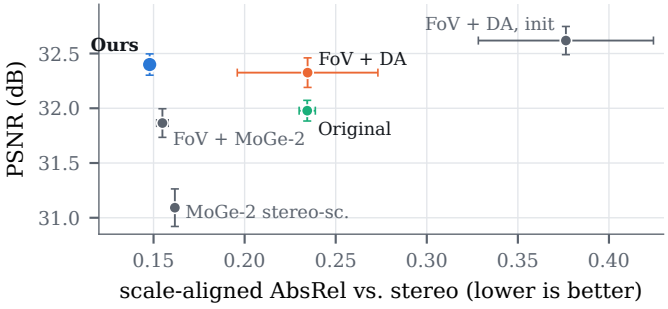


Fig. 6. Image quality vs. geometric accuracy, 3-seed means (±1 s.d.). Ours is the only setup better than the baseline on both.

matched FoV the pose-based scale gives higher PSNR than the stereo fit (31.87 vs. 31.56 dB, 3 seeds vs. 1 seed). PSNR rises as the shared depth/gradient/TV weight λ_d decreases, while the scale-aligned AbsRel stays at 0.148–0.159; disabling this group together with the normal and depth-uncertainty terms gives the best PSNR among the MoGe-2 variants. Because several terms are disabled jointly, this does not identify the effect of each term. Replacing the pseudo-normal loss by MoGe-2 normal supervision, compared at the same stereo-fitted scale, lowers PSNR by 0.12–0.28 dB with AbsRel changes of at most 0.003. BA cuts long-baseline epipolar errors from up to 6.8 to ∼2 px, but its per-frame poses jitter ∼10× more than the kinematic ones and cost about 2 dB; a temporally smoothed correction still trails the robot poses, plausibly because rigid BA absorbs breathing motion (for MoGe-2 the prior scale also changes with the poses).

C. Qualitative results

Fig. 4 shows rendered depth and its error against the stereo reference on a test frame, using the median-error seed of each setup. Both Depth-Anything models render a sharp depth step below the instrument ring that the stereo depth does not support; it appears as a dark band in their error maps. Correcting the field of view alone does not remove it. The MoGe-2 model renders a smoother surface whose scale-aligned error is lower over most of the tissue (AbsRel 0.139 vs. 0.231 and 0.248 on this frame). The maps are aligned per frame, so they show shape agreement, not metric depth.

D. Computational cost

We separate offline preprocessing, training, rendering and the cost of the uncertainty signals. *Preprocessing:* MoGe-2 is larger than Depth-Anything (331 M vs. 97 M parameters) and runs on the GPU in 0.22 s per frame (3.2 GB peak); in our setup Depth-Anything ran on the CPU at 0.75 s per frame, so the two times are not directly comparable. *Training and rendering:* training time (15.9 vs. 16.0 min), the final number of Gaussians (327 k vs. 330 k) and rendering speed (38 FPS) are comparable to the baseline (Table II); we did not measure training memory. *Uncertainty:* prior disagreement needs no additional model training, but at a new frame it needs the monocular prior (0.22 s for MoGe-2, i.e. about 4.5 FPS if run serially) plus a median alignment and per-pixel arithmetic; it is inexpensive only when the prior is already computed, as it is for training frames. The seed ensemble needs three trained models, and ray spread and deformation magnitude need one extra rasterisation pass each.

TABLE V

MONOCULAR PRIORS ALONE ON THE TEST FRAMES, AGAINST SGBM STEREO. RAW SCALE: MEAN PER-FRAME MEDIAN RATIO OF STEREO TO PRIOR DEPTH (CV: ITS COEFFICIENT OF VARIATION OVER FRAMES). ALIGNED ABSREL AND MEDIAN RELATIVE ERROR AFTER PER-FRAME MEDIAN ALIGNMENT; VALID: FRACTION OF PIXELS WITH STEREO AND PRIOR DEPTH.

Seq.	Prior	raw scale	CV	aligned AbsRel	median rel.	valid
P1	Depth-Anything ViT-B	3.450	3.4 %	0.237	0.166	0.51
P1	MoGe-2 ViT-L	0.168	12.8 %	0.148	0.099	0.51
P2_5	Depth-Anything ViT-B	2.946	1.3 %	0.059	0.044	0.60
P2_5	MoGe-2 ViT-L	0.056	6.4 %	0.036	0.028	0.60

TABLE VI

DEPTH ERROR OF THE TRAINED MODELS AGAINST SGBM STEREO UNDER THREE ALIGNMENTS (MEAN OF 3 SEEDS): PER-FRAME MEDIAN ALIGNMENT (ALIGNED), ONE FACTOR FITTED ON THE TRAINING VIEWS (FIXED), AND NONE (METRIC, RAW RENDERED DEPTH IN THE POSE UNITS). SCALE: MEAN PER-TEST-FRAME ALIGNMENT FACTOR AND ITS CV OVER FRAMES.

Setup	aligned	fixed	metric	train scale	test scale	CV
P1						
Original	0.234	0.236	0.744	4.15	4.19	5.4 %
+ FoV fix (DA)	0.235	0.236	0.737	4.12	4.14	4.4 %
+ FoV, MoGe-2 + depth loss	0.155	0.157	0.470	1.87	1.91	5.3 %
Ours	0.148	0.151	0.462	1.83	1.87	5.1 %
P2_5						
Original	0.063	0.063	0.662	3.03	3.04	0.7 %
+ FoV fix (DA)	0.064	0.064	0.662	3.04	3.04	0.6 %
+ FoV, MoGe-2 init, unscaled	0.046	0.046	15.91	0.060	0.060	0.6 %
+ FoV, MoGe-2 init, stereo-scaled	0.046	0.046	0.056	1.04	1.04	0.8 %

E. Prior accuracy, metric depth and scale drift

Two questions underlie the main metric: how much does per-frame alignment hide, and how much of the reconstructed geometry comes from the prior? Table VI shows that one scale factor fitted on the training views gives almost the same error as per-frame alignment (at most 0.003 higher), and that the per-frame factors vary by only 0.4–6.2 % over the test frames in individual runs, so per-frame alignment hides little temporal scale drift in these clips. It does hide the absolute scale. On P1 the raw rendered depth of our model must be enlarged by 1.87 to match the stereo reference (metric AbsRel 0.462), and the DA models by about 4.2 (0.74). The pose-based scale thus does not yield metric agreement with the stereo reference; the known-pose estimate and the stereo fit differ by the same factor (0.093 vs. 0.165; Section III-D). On P2_5 the stereo-scaled model is close to metric (factor 1.04, metric AbsRel 0.056), as expected from a stereo calibration, while the unscaled MoGe-2 model is 17× off.

Table V shows that the reconstructions largely inherit the shape accuracy of their priors. On P1 the aligned AbsRel of the prior alone (0.237 DA, 0.148 MoGe-2) matches that of the trained models (0.234–0.235 and 0.148), and on P2_5 training

TABLE VII

SECOND SEQUENCE: STEREOMIS P2_5, FRAMES 5301–5499 (EVERY SECOND FRAME; 100 FRAMES, 13 TEST VIEWS; MEAN ± SAMPLE S.D., 3 SEEDS). THE CAMERA IS STATIC, SO THE KNOWN-POSE SCALE IS UNDEFINED; THE SCALED ROW USES A SEPARATE, STEREO-ASSISTED PROTOCOL. ABSREL: PER-FRAME SCALE-ALIGNED, AGAINST STEREO.

Setup	PSNR↑	SSIM↑	LPIPS↓	AbsRel↓	δ<1.25 ↑
Monocular inputs (left images, known poses)					
Original, reproduced	26.70±.10	0.729±.003	0.184±.003	0.063±.001	0.967±.002
+ FoV fix (DA)	27.89±.03	0.778±.002	0.168±.002	0.064±.001	0.965±.002
+ FoV, MoGe-2 init, unscaled	27.95±.02	0.776±.001	0.201±.001	**0.046**±.001	**0.996**±.001
Stereo-assisted scale calibration (training frames)					
+ FoV, MoGe-2 init, scaled†	**28.66**±.08	**0.807**±.003	**0.153**±.007	**0.046**±.001	**0.996**±.003

† Scale ×0.0572, the median over training frames 5301, 5321, …, 5481 (every 10th frame; none is a test frame) of the per-frame median ratio of stereo to MoGe-2 depth. Bold AbsRel values tie at the displayed precision (unrounded: 0.0459 unscaled, 0.0458 scaled).

even increases it (0.036 to 0.046 for MoGe-2, 0.059 to 0.064 for DA). Most of the geometric gain therefore comes from choosing a better prior and keeping it as the initial surface, not from the optimisation improving on it. MoGe-2 also varies more in scale from frame to frame than DA (CV 12.8 vs. 3.4 % on P1), which a single global factor does not correct.

F. Second sequence

To test whether the FoV and initialisation findings transfer, we repeat the main configurations on StereoMIS P2_5, frames 5301–5499 at every second frame, a clip also used by Endo-PairGS [15]. Its rectified focal length is 559.9 px (FoV 59.5°), and the released-style preprocessing again stores the unrectified 1033.9 px. The camera is essentially static: the largest distance between any two camera centres is 0.12 mm, so triangulation has no parallax and the scale of Section III-D cannot be estimated. We therefore report MoGe-2 initialisation unscaled, which uses only monocular inputs, and, as a separate stereo-assisted protocol, with a scale fitted to stereo on training frames (×0.0572; MoGe-2 depth is 17.8× too large on the test frames, Table V).

Table VII reproduces the FoV and initialisation findings of P1. The FoV fix adds 1.19 dB PSNR without changing the scale-aligned depth error (0.064 vs. 0.063), and MoGe-2 initialisation reduces it by 28 % relative to FoV-corrected DA (0.064 to 0.046) and raises δ<1.25 to 0.996, with or without scale correction. Without scaling, PSNR and SSIM match FoV-corrected DA but LPIPS is worse (0.201); the stereo-fitted scale gives the best values on all three image metrics, 1.96 dB PSNR above the reproduced baseline. We did not run the objective, normal-loss or pose ablations on P2_5, so the conclusions about depth supervision and poses rest on P1 alone. Endo-PairGS reports 29.06 dB for Endo-4DGS on this clip [15], 2.4 dB above our reproduction; we cannot reconcile this from the published protocol description and compare only within our protocol.

VI. EXPLAINABILITY: WHERE DOES THE DEPTH DISAGREE?

Image metrics say how well a model re-renders the video, not where its depth is wrong. We now ask which per-pixel signals of a trained model rank its depth error against the stereo reference (Section III-K), and in which regions the error lies. All signals are computed from the model and its monocular prior before any stereo mask is applied. Table VIII summarises 2 sequences × 2 setups × 2 stereo references, each averaged over the test frames and three seeds.

Accumulated opacity ranks depth errors weakly. The accumulated opacity that Endo-4DGS uses as its confidence map reaches $\rho \leq 0.12$ and -1 to 17 % of the attainable ranking gain over all eight conditions, and is no better than a random ordering for both Depth-Anything models (-1 to 4 %). For our model it gives a small positive gain (0–17 %; Fig. 8), and the error in its most uncertain decile is 0.9–2.0× that of its least uncertain decile (Fig. 9). This is consistent with how it is trained: the confidence loss uses global reductions (Section III-G), so $\hat{w}$ is not optimised as a per-pixel estimate of depth error. Our finding concerns this signal and loss, not learned uncertainty in general.

Disagreement with the monocular prior is the strongest tested individual signal. Measuring how far the rendered depth departs from the median-aligned prior of the same frame gives the highest ρ and gain in seven of the eight conditions ($\rho = 0.28$–0.55, 34–65 % of the attainable gain); the exception is FoV-corrected DA on P1 with the standard reference, where the rank combination is better (49 vs. 39 %). Pixels in its most uncertain decile have 1.9–4.8× the error of those in its least uncertain decile. With the strict reference its gain increases in three of the four settings, so it persists after filtering some stereo mismatches; this does not exclude errors shared by the reconstruction and the stereo reference. The signal is relative: it ranks pixels within a frame but does not provide an absolute reliability threshold. A plausible reason for its strength is that the reconstruction's depth largely inherits the prior's (Table V), so where training pulls the depth away from the prior, one of the two is likely wrong. The seed ensemble is a reasonable second for Depth-Anything (25–37 %) but inconsistent for MoGe-2 initialisation, whose seeds agree closely (-10 to 31 %). Ray depth spread is inconsistent (-13 to 26 %), and deformation magnitude is informative mainly on the static-camera sequence (26–39 % for ours on P2_5), where apparent motion reflects tissue motion alone. The rank combination does not improve on prior disagreement elsewhere.

Where the error lies. Fig. 10 splits the error by disjoint region. On P1, depth edges (8.6 % of the scored pixels) have the highest per-pixel error for every model with the standard reference, but most of the total error lies on ordinary tissue: for our model, tissue carries 63 % of the summed error, depth edges 22 %, specular highlights 8 % and the ring 7 % (computed from per-frame error sums). Relative to FoV-corrected DA, MoGe-2 initialisation reduces the per-pixel error most on the ring (-46 %, 0.320 to 0.172) and on tissue (-42 %) and least at depth edges (-15 %). On P2_5, which has no ring and smaller errors overall, tissue carries 84 % of the error, and the reduction relative to FoV-corrected DA is largest at depth edges (-32 %) and on tissue (-29 %) and smallest at specular highlights (-16 %). The strict reference

TABLE VIII

WHICH SIGNAL RANKS THE STEREO-REFERENCE-RELATIVE DEPTH ERROR? SPEARMAN ρ BETWEEN EACH PER-PIXEL SIGNAL (COMPUTED FROM THE MODEL AND ITS MONOCULAR PRIOR ONLY) AND THE PER-FRAME SCALE-ALIGNED ABSREL AGAINST SGBM STEREO, AND THE SHARE OF THE ATTAINABLE RANKING GAIN G (EQ. 9; RATIO OF THE MEAN AURG TO THE MEAN RANDOM-ORDER AUSE OVER 25 (P1) OR 13 (P2_5) TEST FRAMES AND THREE SEEDS; 100 % = ORACLE, 0 % = RANDOM). STRICT: STEREO PIXELS THAT PASS A LEFT–RIGHT CONSISTENCY CHECK. BEST PER COLUMN IN BOLD.

	P1, FoV + DA				P1, ours				P2_5, FoV + DA				P2_5, ours (stereo-scaled)			
	SGBM		strict		SGBM		strict		SGBM		strict		SGBM		strict	
Signal	ρ	gain	ρ	gain	ρ	gain	ρ	gain	ρ	gain	ρ	gain	ρ	gain	ρ	gain
accumulated opacity (Endo-4DGS confidence)	-0.00	-1	0.01	-1	0.11	7	0.04	0	0.04	-1	0.04	4	0.06	3	0.12	17
ray depth spread	0.20	20	0.21	10	0.15	5	0.09	-4	0.07	-13	0.04	12	0.14	16	0.18	26
deformation magnitude	-0.06	-2	-0.01	6	0.04	9	0.03	18	0.10	7	0.07	-1	0.25	39	0.17	26
seed ensemble (3 seeds)	0.24	31	0.22	37	0.15	8	0.06	-10	0.29	29	0.21	25	0.20	31	0.21	24
prior disagreement	0.30	39	**0.55**	**65**	**0.36**	**35**	**0.28**	**43**	**0.40**	**49**	**0.33**	**34**	**0.33**	**47**	**0.40**	**48**
combined (rank mean)	**0.39**	**49**	0.52	63	0.30	31	0.21	27	0.36	41	0.28	33	0.30	42	0.35	43

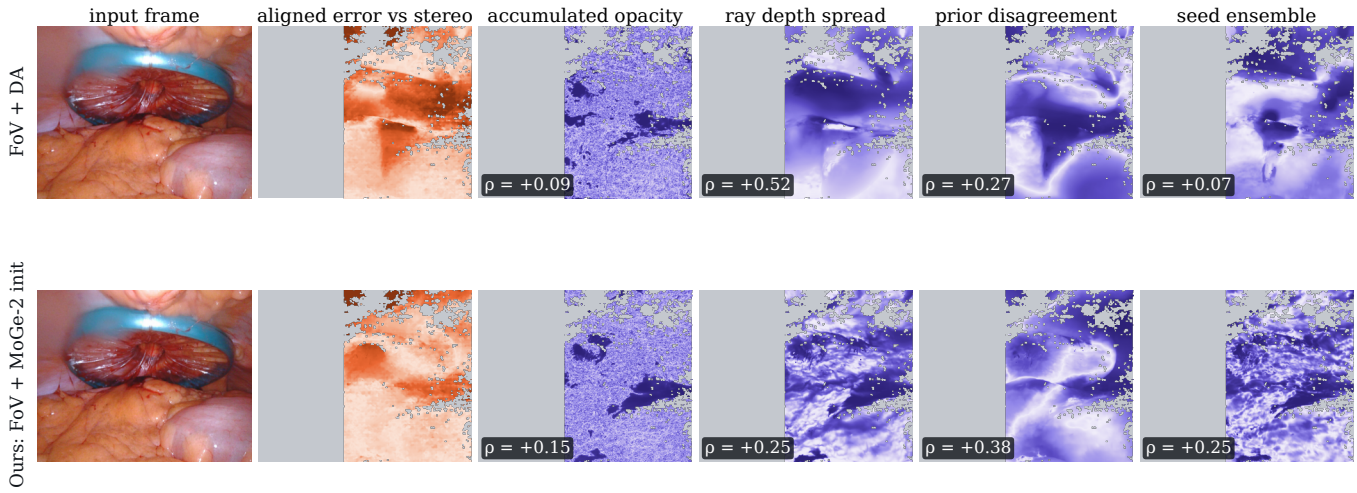


Fig. 7. Per-frame scale-aligned depth error against the stereo reference and four uncertainty signals for test frame 898 of P1 (seed 0). Uncertainty maps show per-frame percentile ranks (dark = most uncertain); ρ is the Spearman correlation with the error on this frame. Grey: no stereo reference. Accumulated opacity is nearly unrelated to the error here, whereas prior disagreement and, for Depth-Anything, ray depth spread highlight the erroneous band below the instrument ring.

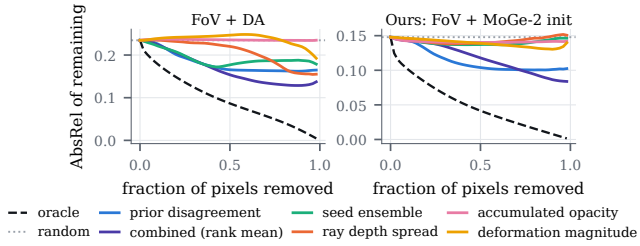


Fig. 8. Sparsification curves on P1: mean scale-aligned AbsRel of the pixels that remain after removing the most uncertain fraction. Curves below the random line rank error; the dashed oracle removes pixels by their true error.

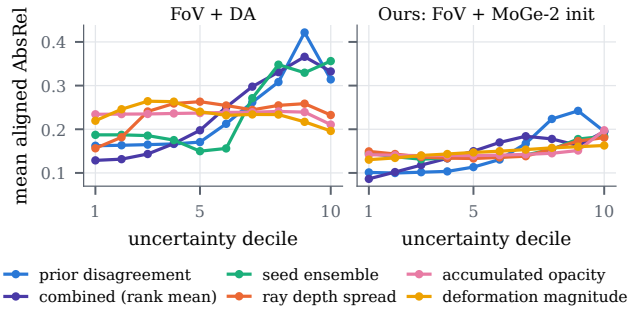


Fig. 9. Error by uncertainty decile on P1, a ranking diagnostic (not a calibration of predicted error magnitudes). A useful signal rises from left to right.

changes which pixels are scored (on P1 specular pixels rise from 4.2 to 5.9 % and ring pixels fall from 5.6 to 3.7 % of the scored pixels) and raises the per-pixel error of specular and ring regions for all models, but tissue still carries 56 % of our error. The regions are defined by colour and stereo-gradient rules, so these numbers indicate trends rather than precise per-class errors.

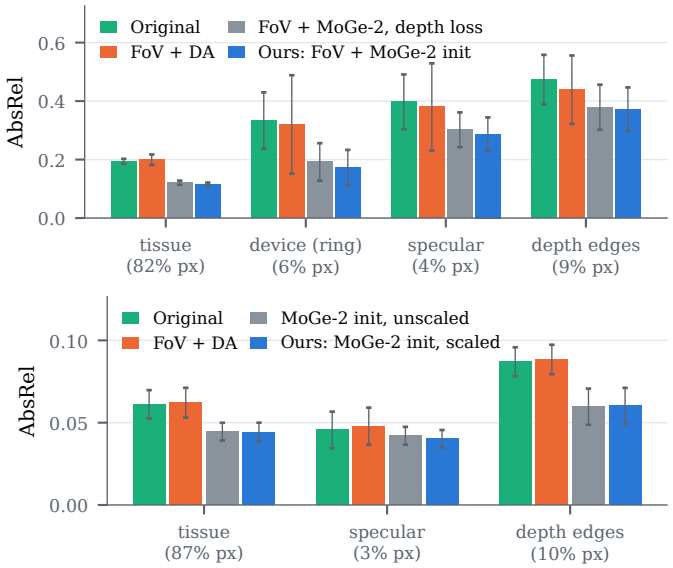


Fig. 10. Per-frame scale-aligned AbsRel by disjoint region on P1 (top) and P2_5 (bottom), mean ± s.d. over test frames and seeds; the share of scored pixels per region is given below each label.

TABLE IX
PUBLISHED RESULTS ON STEREOMIS P1 FRAMES ∼800–1000, ONE ROW PER SOURCE, AS PRINTED (PROTOCOL-DEPENDENT CONTEXT, NOT A CONTROLLED COMPARISON); OURS: MEAN OF 3 SEEDS. FPS AND GPU BELONG TO THE SAME SOURCE. ABSREL (PER-FRAME SCALE-ALIGNED, VS. STEREO) IS REPORTED ONLY BY THIS WORK.

Method	Source	PSNR↑	AbsRel↓	FPS	GPU
Endo-4DGS	own [12]	32.69	–	100	RTX 4090
Endo-4DGS	[15]	32.17	–	–	A100
Endo-4DGS	[16]	32.19	–	100	RTX 4090
ST-Endo4DGS	[16]	32.90	–	123	RTX 4090
EndoGaussian	[15]	32.79	–	–	A100
SurgicalGS	[15]	32.98	–	–	A100
SurgicalGS	[17]	31.60	–	183	RTX 4090
EndoWave	own [17]	33.26	–	77	RTX 4090
Deform3DGS	[15]	33.35	–	–	A100
Endo-PairGS	own [15]	33.56	–	100	A100
Endo-G^{2}T	own [16]	33.58	–	148*	RTX 4090
Endo-TSR†	own [25]	38.45	–	–	–
Endo-4DGS, reproduced	this work	31.98	0.234	38	RTX 4060 Ti
Ours (×0.093)	this work	32.40	**0.148**	38	RTX 4060 Ti
Ours, held-out-clean scale (×0.104)	this work	32.29	0.147	–	RTX 4060 Ti

†Uses stereo depth. *Rasterisation only, as stated by the source. Our FPS: 640×512.

VII. COMPARISON WITH THE STATE OF THE ART AND NOVELTY

Table I (Section II) and Table IX place our work against the literature. Controlled comparisons are only those within this study; published numbers from other papers are context. We support each claim with evidence and state where the evidence ends.

Claim 1: higher PSNR and lower scale-aligned depth error than the method we build on. Under an identical protocol (same clip, split, metrics, hardware and three seeds), our model improves on reproduced Endo-4DGS on P1 by +0.42 dB PSNR (per seed +0.44, +0.45, +0.37) and reduces the per-frame scale-aligned AbsRel by 37 %, with lower error on all 25 test frames, while SSIM changes by only +0.001 and LPIPS worsens from 0.178 to 0.189. With the held-out-clean scale factor the result is 32.29 dB PSNR and 0.147 AbsRel over three seeds (+0.31 dB and −37 % relative to the reproduced baseline; SSIM −0.001, LPIPS 0.194), so the depth gain does not depend on the test frames, while the PSNR gain is 0.11 dB smaller. Its PSNR exceeds the Endo-4DGS values reported by two independent re-implementations on this clip (32.17 and 32.19 dB [15], [16]), but not the 32.69 dB printed by its authors [12]; these are cross-protocol numbers.

Claim 2: a depth evaluation on this clip. Among the works we reviewed, none of the methods evaluated on P1 frames 800–1000 [12], [15]–[17], [25] reports depth or surface accuracy (Table IX). Our scale-aligned AbsRel of 0.148 measures shape agreement with an estimated stereo reference, not verified surface accuracy, and our P1 reconstructions are not metric (Section V-E). Local-EndoGS reports a lower AbsRel (0.112 and 0.129) with a similar metric [33], but on different and longer sequences with its own reference depth, so the two are not comparable.

Claim 3: components and findings. Known-pose triangulation and median scale fitting are established ingredients; our contribution is their controlled use and evaluation in this endoscopic setting. Among the works we reviewed, we found none that (i) uses MoGe-2 as the geometry prior for endoscopic Gaussian splatting and measures that its metric scale is 6–18× off on endoscopic images; (ii) scales a metric prior by known-pose triangulation, which, unlike per-frame fitting to rendered depth [33] or pseudo-metric optimisation [34], needs neither stereo nor a first reconstruction (it converts units consistently with the poses but does not remove the prior's frame-to-frame scale variation); (iii) reports the field-of-view error in the released StereoMIS preprocessing of Endo-4DGS, whose StereoMIS protocol several later works adopt [14], [16], [17] (whether their preprocessing inherits the error is not stated); or (iv) shows that, with such an initialisation, removing the depth-related losses as a group improves PSNR at equal scale-aligned depth error, and that a model with the highest PSNR can have substantially worse depth agreement.

Claim 4: explaining geometric error. Among the works we reviewed, none evaluates per-pixel geometric uncertainty of deformable Gaussian splatting in endoscopy against an independent depth reference (Section II-E). Endo-NeRF++ [43] evaluates colour uncertainty of a dynamic NeRF against image error, and depth-AUSE benchmarks exist for static, non-medical scenes [36]–[39]. Our benchmark (Section VI) shows on two sequences and two stereo references that the accumulated opacity used as confidence by Endo-4DGS ranks depth errors weakly (−1 to 17 % of the attainable gain) and that prior disagreement is the strongest tested individual signal (34–65 %), with no additional training once the prior is available. Per-ray depth variance [38] and ensembles [41] are established signals; we claim only their evaluation in this setting.

Where we do not lead. On this clip, recent methods with stronger representations or priors report 0.86–1.18 dB higher

PSNR (Endo-PairGS, Endo-G^{2}T, Deform3DGS as re-run by Endo-PairGS, EndoWave), and Endo-TSR, which uses stereo depth, reports 38.45 dB. These are published numbers under protocols that may differ from ours, so they are context rather than a controlled ranking; we do not claim state-of-the-art image quality. Our contributions concern the depth prior, its scale and the training objective; whether they combine with these architectures is untested, and a controlled comparison including a geometric evaluation of those methods is the natural next step.

VIII. Conclusion

We revisited monocular 4D Gaussian splatting for endoscopy and measured not only how well it re-renders the video but how well its depth agrees with a stereo reference and where it disagrees. On StereoMIS P1, corrected intrinsics and pose-scaled MoGe-2 initialisation improve PSNR from 31.98 to 32.40 dB and reduce the per-frame scale-aligned depth AbsRel against SGBM stereo from 0.234 to 0.148, averaged over three seeds, while LPIPS increases from 0.178 to 0.189; with the scale factor estimated from training frames only and filtered for consistency with the known poses, the result is 32.29 dB and 0.147. On the static P2_5 clip, MoGe-2 initialisation reduces the aligned AbsRel from 0.064 to 0.046 with monocular inputs, and a separate stereo-calibrated protocol improves PSNR from 26.70 to 28.66 dB. These measurements assess reference-relative depth shape; our P1 reconstructions are not metric (raw depth about 1.9$\times$ too shallow relative to stereo), and the scale of a state-of-the-art point-map model does not transfer to endoscopic images (6–18$\times$ off). Image-metric rankings need not match depth-accuracy rankings on these clips. Among the tested per-pixel signals, accumulated opacity ranks depth errors weakly, whereas disagreement with the monocular prior is the strongest individual signal on both sequences and under a strict left–right-consistent reference. Depth edges have the highest per-pixel error, but most of the total error lies on ordinary tissue.

Limitations. The study covers two short StereoMIS clips and evaluates only pixels with a usable stereo reference, which remains an estimate even after left–right filtering. Per-frame alignment hides global scale errors, which we report separately. The known-pose scale needs camera motion and quasi-static correspondences and could not be used on P2_5. The objective, normal-loss and pose ablations are on P1 only, and the jointly disabled depth terms are not separated. EndoNeRF data were not available to us, and comparisons with other methods rely on published numbers under possibly different protocols. **Future work** will test additional sequences and independently measured geometry, retrain the strongest open-source methods under the same protocol, separate the individual loss terms, and test whether prior disagreement can guide adaptive refinement.

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